

Question 1: Prove Theorem 3.1.3: A non-trivial connected graph G has an eulerian circuit if and only if every vertex of G has even degree

($\implies$) Assume that G has an Eulerian circuit $v_0, v_1, \dots, v_k, v_0$. For every $1 \leq i \leq k$, the occurrence of a vertex v_i involves adding 2 edges to its degree, namely $v_{i-1}v_i$ and v_iv_{i+1} . Since every edge is unique, we will always add 2. For vertex v_0 , we have edges v_0v_1 and v_kv_0 , an even amount. After this, every edge has been counted and it follows every vertex has even degree.

($\impliedby$) We strong induct on the size of $G = m$. The case when $m = 2$ is trivial. (bounce back and forth between the vertices using each edge once). [Or we can use the trivial graph as the base case for $m = 0$] Assume we can find a Eulerian circuit for any graph of size less than m . Now let G be a graph of size m . If G is acyclic (since it is connected), it is a tree and thus has at least 2 end-vertices, a contradiction. So G is cyclic and has a cycle $C : w_1w_2w_3\dots w_k$. Consider the components after cutting edges $\{w_1w_2, w_2w_3, \dots, w_kw_1\}$ being $C_1, \dots, C_N$. Each of these components has size less than m , and even degrees (as only the cycle vertices are affected, each removing 2 degrees) and are connected to cycle C by some vertex w_i . By the inductive hypothesis, so have Eulerian circuits P_i . We construct a Eulerian path as follows: Proceed along cycle C . If I come across some vertex on C in C_i , use path P_i unless that path has been processed already, then just move to the next vertex on C . This new circuit is Eulerian as required.

Question 2: Prove Theorem 3.1.4: A non-trivial connected G has an Eulerian trail if and only if G contains exactly two vertices of odd degree. Furthermore, every Eulerian trail begins at one of these odd vertices and ends at the other

We can prove it similarly as to how we did in Question 1, but a shorter way is to modify graph G into H by adding one edge between these odd vertices (call them u and v). Then G has exactly 2 odd vertices if and only if H has even vertices if and only if H has a Eulerian circuit. Now if $P : v_0\dots v_{i-1}uvv_i\dots v_kv_0$ is a Eulerian circuit for H , then $vv_i\dots v_kv_0v_1\dots v_{i-1}u$ is an Eulerian trail (in G) starting and ending at these odd vertices. On the other hand, if $uv_0v_1\dots v_{i-1}v_i\dots v_kv$ is an Eulerian trail starting and ending at these odd vertices (in G), then $vv_0v_1\dots v_kv$ is an Eulerian circuit in H . So We have an Eulerian trail in G starting and ending at these odd vertices if and only if we have an Eulerian circuit in H . Finally, we get G has an Eulerian trail starting and ending at these odd vertices if and only exactly 2 vertices have odd degree. Clearly Every Eulerian trail in G must start and end at these odd vertices, by the degree argument above if they start somewhere else these odd vertices have even degree.

Question 3: Prove that the converse of Theorem 3.2.3 is not true: Show that for a graph G , if every proper non-empty subset $S \subseteq V(G)$, we have $k(G - S) \leq |S|$, then G is Hamiltonian is false.

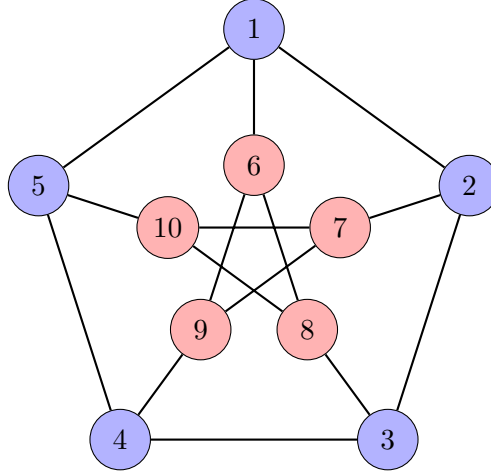


Figure 1: The Petersen Graph

The Petersen Graph G is not Hamiltonian and $\kappa(G) = 3$ so for $|S| < 3$ clearly $k(G - S) \leq |S|$. If we think of how to split G into 3 components, it involves removing 5 vertices (isolating 2 outer vertices). This is clearly the minimum since for any $u, v, w \in V(G)$, $p(u, v) = p(v, w) = 3$ and there is only one $u - w$ path passing through v . (Each path must remove a vertex to separate them into 3 components, with only one vertex being shared along other paths). Creating 4 components is impossible since if we want 4 vertices in different components, pick one vertex. This has 3 neighbours that can't be the other vertices. This leaves 6 other vertices. Picking another vertex also has 3 neighbours, with at most 1 shared with vertex 1. This leaves 3 vertices for 2 other selections, impossible as each vertex will guarantee at least 2 more vertices we cannot pick. A simpler graph is simply the disconnected graph with 2 vertices.

Question 4: Prove Corollary 3.2.5: Let u and v be distinct non-adjacent vertices of a graph G of order n such that $\deg u + \deg v \geq n$. Then G is Hamiltonian if and only if $G + uv$ is Hamiltonian

Clearly if G is Hamiltonian, then so is $G + uv$ (the Hamiltonian cycle for G is also a Hamiltonian cycle for $G + uv$). So suppose that $G + uv$ is Hamiltonian. Let $P : v_0 v_1 \dots v_k v_0$ be a Hamiltonian cycle. If edge uv is not used, we are done and this is a Hamiltonian cycle for G . Otherwise $P : v_0 v_1 \dots v_i u v v_{i+1} \dots v_0$ means $vv_{i+1} \dots v_k v_0 v_1 \dots v_i u := vv_0 w_1 \dots w_{k-2} u$ is a Hamiltonian path. Since $\deg u + \deg v \geq n$, and there are $n - 2$ other vertices, there are at least two vertices w_i and w_j neighbouring both (and all other vertices neighbour one). Without loss of generality, assume $i < j$. First suppose $i + 1 \neq j$. If w_{i+1} neighbours v , we get Hamiltonian cycle $w_j \dots w_{i+1} v w_0 \dots w_i u w_{k-2} \dots w_j$. If w_{j-1} neighbours u , we get Hamiltonian cycle

$w_i \dots w_{j-1} u w_{k-2} \dots w_j v w_0 \dots w_i$. So assuming neither of those cases hold, we can look at whether w_{i+2} neighbours v or w_{j-2} neighbours u . Again, in these cases we find Hamiltonian cycles similarly. Again, assuming these don't happen, we can continue working inwards and we will eventually find a pair that satisfies this (otherwise we get a contradiction when $j - N < i + N$). What's left is to check the case when $j = i + 1$. This is simple, we use Hamiltonian cycle $vw_0 \dots w_i u w_{k-2} \dots w_j v$.

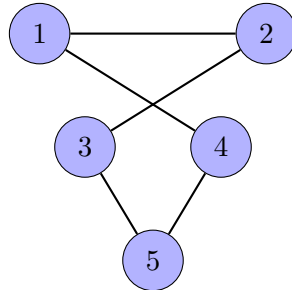
Using the fact it is a Corollary, we follow a similar proof that they did. Clearly if G is Hamiltonian, then so is $G + uv$. Suppose there is a graph G that is not Hamiltonian, but $G + uv$ is. Let C be a Hamiltonian cycle of $G + uv$. Necessarily, the edge uv lies on C , so there is a $u-v$ path $P : u = u_1, u_2, \dots, u_n = v$ that contains every vertex of G . If $u_1 u_i \in E(G)$, then necessarily $u_{i-1} u_n \notin E(G)$, for otherwise $u_1, u_2, \dots, u_{i-1} u_n, u_{n-1}, \dots, u_i, u_1$ is a Hamiltonian cycle of G . Consequently, $\deg(u_n) \leq n-1 - \deg u_1$. So $\deg u + \deg v \leq n-1$, a contradiction.

Question 5: Let G_1 and G_2 be two Eulerian graphs with no vertex in common. Let $v_1 \in V(G_1)$ and $v_2 \in V(G_2)$. Let G be the graph obtained from $G_1 \cup G_2$ by adding the edge $v_1 v_2$. What can be said about G ?

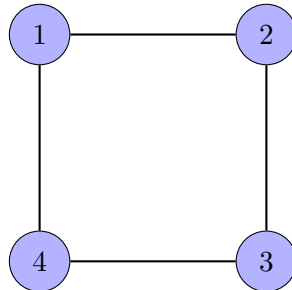
G has an Eulerian trail. In G_1 and G_2 , every vertex has even degree (from the graphs being Eulerian). Adding edge $v_1 v_2$ in G means v_1 and v_2 have odd degrees. G_1 and G_2 share no vertices, so only v_1 and v_2 have odd degrees and thus G has an Eulerian trail.

Question 6: Give an example of a graph G such that:

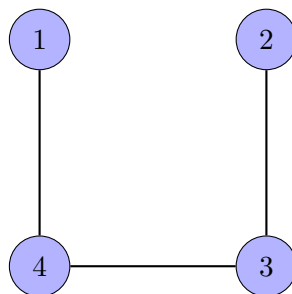
(a) both G and $\bar{G}$ are Eulerian



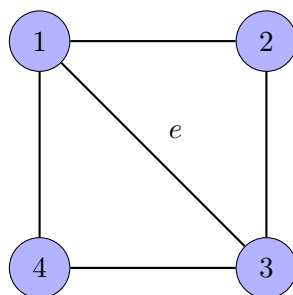
(b) G is Eulerian but $\bar{G}$ is not



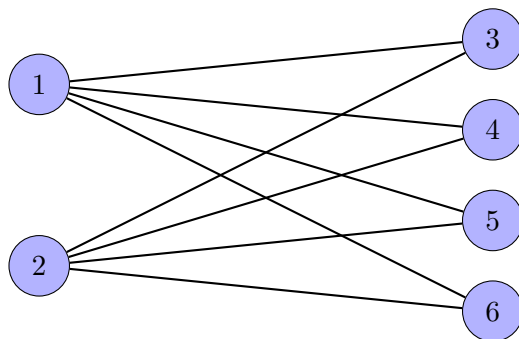
(c) both G and $\bar{G}$ contain an Eulerian trail



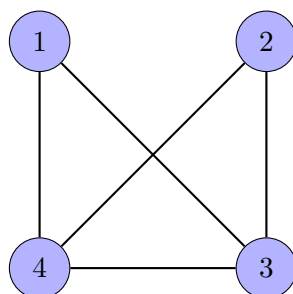
(d) G contains an **Eulerian trail** and an edge e such that $G - e$ is **Eulerian**



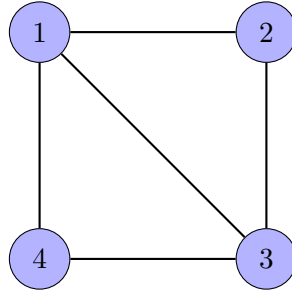
(e) G is **Eulerian** but not **Hamiltonian**



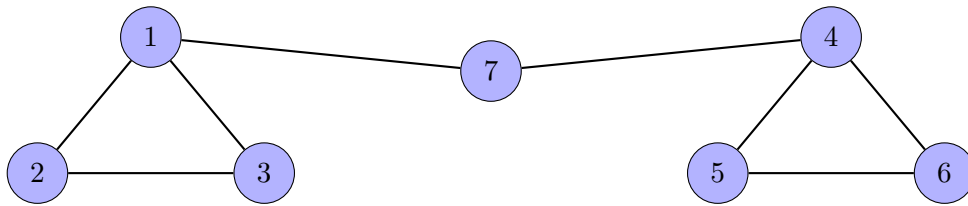
(f) G is **Hamiltonian** but not **Eulerian**



(g) G is **Hamiltonian** and has an **Eulerian trail**



(h) G is **not Hamiltonian** and G has an **Eulerian trail**

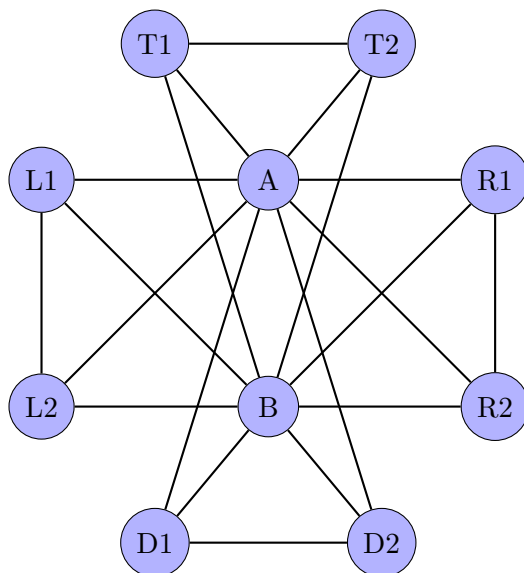


Question 7: Show that every nontrivial connected graph has a closed spanning walk that contains every edge exactly twice

Modify our graph G into a non-simple graph H , where every edge is duplicated once. Then H has even degree and thus has a Eulerian circuit. This is equivalent to G having a closed spanning walk that contains every edge exactly twice.

Question 8: Prove or disprove:**(a) If G is 2-connected, then G is Hamiltonian**

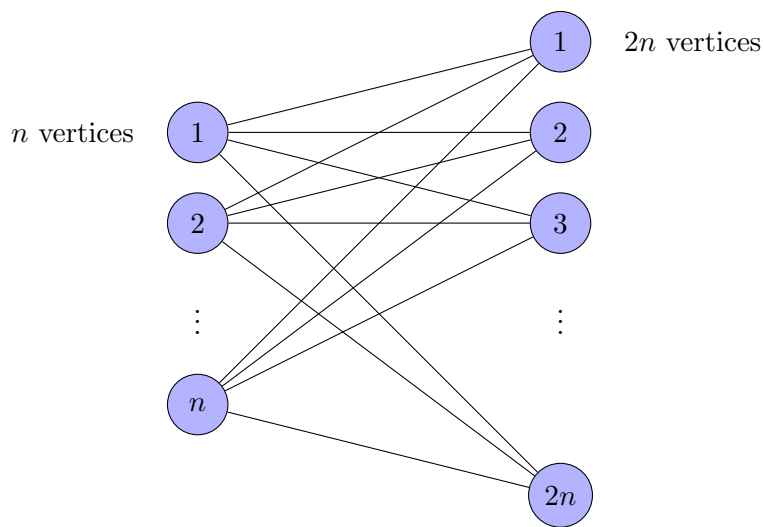
This is not true. Consider graph: (or Petersen graph)



This graph is 2-connected ($\kappa(G) = 2$ for vertex-cut $S = \{A, B\}$) but not Hamiltonian, since $k(G - S) = 4$ but $|S| = 2$ (or trying to get a Hamiltonian cycle, the K_3 instances force visiting the same vertex twice)

(b) There exists an integer k such that every k -connected graph is Hamiltonian

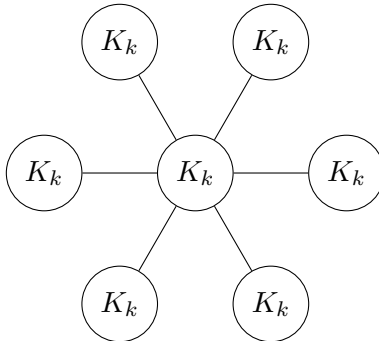
This is false. A counterexample is $K_{n,2n}$. The connectivity $\kappa(K_{n,2n}) = \min\{n, 2n\} = n$. We show that $K_{n,2n}$ is not Hamiltonian:



Call the left partite set V_1 and the right partite set V_2 and assume we have a Hamiltonian cycle P :

$u_1, u_2, \dots, u_{3n}, u_1$, then notice that every edge $u_i u_{i+1}$ must have one vertex in each partite set. But then for the $2n$ vertices in V_2 , to each occur exactly once in P , there must be $2n$ vertices in V_1 . But there is only n vertices, so we cannot have a Hamiltonian path

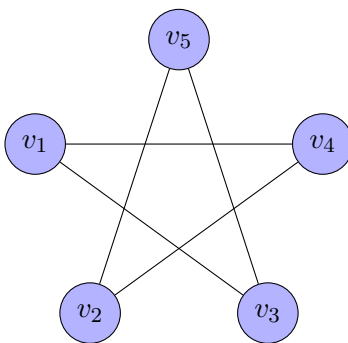
Another counterexample is (using the cutting vertex component theorem):



With $k + 1$ outer components, and every vertex in the center connects to every other in each component

Question 9: Prove that $\bar{C}_n$ is Hamiltonian for $n \geq 5$

In C_n , each vertex has degree 2. In $\bar{C}_n$, every vertex has degree $(n - 1) - 2 = n - 3$. For any non-adjacent vertices u and v , we have that $\deg v + \deg u = 2n - 6 \geq n \iff n \geq 6$. It follows $\bar{C}_n$ is Hamiltonian for $n \geq 6$. The case $n = 5$ is Hamiltonian with cycle $v_1, v_4, v_2, v_5, v_3, v_1$



Question 10: Let G be a 6-regular graph of order 10 and let $u, v \in V(G)$. Prove that G , $G - u$, and $G - u - v$ are all Hamiltonian

Let x, y be nonadjacent vertices in:

- G . Then $\deg x + \deg y = 6 + 6 = 12 \geq 10$ so G is Hamiltonian.
- $G - u$. Then $\deg x + \deg y \geq 5 + 5 = 10 \geq 9$ so $G - u$ (a graph of order 9) is Hamiltonian (each vertex loses at most 1 edge)

- $G - u - v$. Then $\deg x + \deg y \geq 4 + 4 = 8$ so $G - u - v$ (a graph of order 8) is Hamiltonian (each vertex loses at most 2 edges)

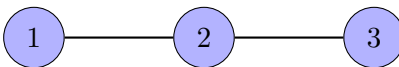
Question 11: The subdivision graph $S(G)$ of a graph G is the graph obtained from G by deleting every edge uv of G and replacing it by a vertex w of degree 2 that is joined to u and v (e.g., the subdivision graph of C_3 is C_6). Prove or disprove: If $S(G)$ is Hamiltonian, then G is Eulerian

This is true. Let G be a graph of order n . If $uv \in E(G)$, we will denote x_{uv} as the new vertex in $S(G)$ such that ux_{uv} and $x_{uv}v \in E(S(G))$. If $S(G)$ is Hamiltonian, we can find a Hamiltonian cycle $P : v_1, x_{v_1v_2}, v_2, x_{v_2v_3}, v_3, \dots, v_n, x_{v_nv_1}, v_1$ (since in $S(G)$, the only neighbours of a vertex $v \in G$ are the new x_{uv} vertices for $u \in N(v)$ in G). Since this is Hamiltonian, every x_{uv} is in the path. It follows that the path $v_1, v_2, \dots, v_n, v_1$ contains every edge. Since every vertex occurs exactly once, every edge in G occurs exactly once (there is a bijection between the x vertices and edges). So P is an Eulerian circuit making G Eulerian

Question 12: A hamiltonian path is a path that contains every vertex of G , and a graph is traceable if it contains a hamiltonian path

- (a) Clearly, every hamiltonian graph is traceable. Is the converse true?

No. Consider graph:



- (b) Let G be a graph of order $n \geq 3$ such that $\deg u + \deg v \geq n - 1$ for every two nonadjacent vertices u, v of G . Prove that G is traceable

Consider modifying graph G into H by adding a vertex x and adding edges ux for every $u \in V(G)$. Then H is a graph of order $n + 1$, and for every nonadjacent vertices u and v , $\deg u + \deg v \geq n - 1 + 2 = n + 1$. So H is Hamiltonian with cycle $P : v_1, v_2, \dots, v_i, x, v_{i+1}, \dots, v_n, v_1$. Then the path $v_{i+1}, \dots, v_n, v_1, v_2, \dots, v_i$ is a Hamiltonian path for G . So G is traceable.

Question 13: Let G be a graph of order $n \geq 3$ having the property that for each $v \in V(G)$, there is a Hamiltonian path with initial vertex v . Show that G is 2-connected but not necessarily Hamiltonian

For every vertex v in G , there is a path P_v visiting every vertex, starting at v . Then starting at the next vertex is a path visiting every vertex in $G - v$, so $G - v$ is connected and thus G is 2-connected. This graph need not be Hamiltonian, for example,

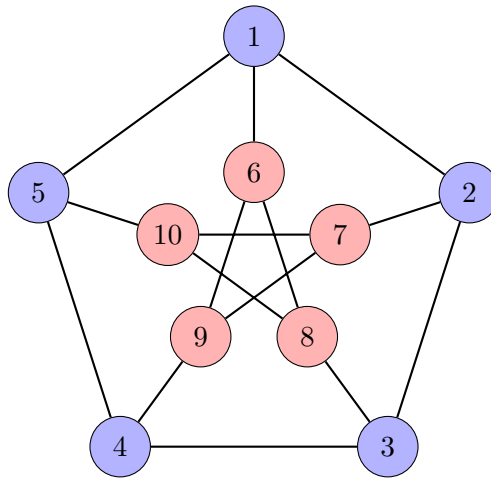


Figure 2: The Petersen Graph

Question 14: If G is a graph of order $n \geq 3$ with $\delta(G) \geq n/2$, prove that G is Hamiltonian

Let u and v be nonadjacent vertices. Then $\deg u + \deg v \geq \delta(G) + \delta(G) \geq n$. So G is Hamiltonian